

Semantic Engine for Multimodal Search

Towards easier data discovery

Data sharing in privacy

Data governance and policy

Project team: Mehrdad Almasi, Tugce Karatas, Pawel Kaminski, Frederic Maguin

One problem two solutions



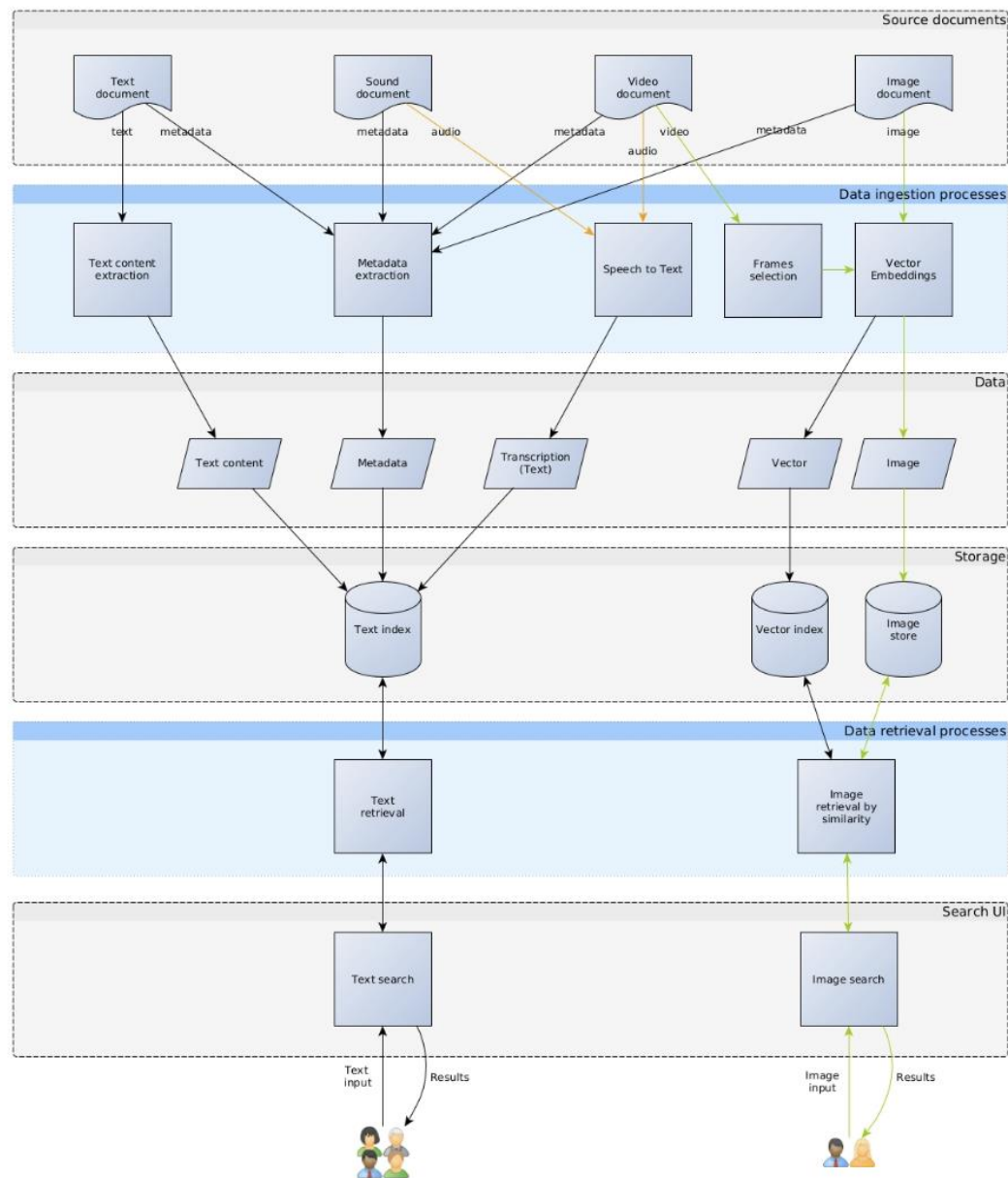
MuSE

<http://muse-test.c2dh.uni.lux:8080>



Search across C2DH publications, datasets and more...

What are you looking for ?



MuSE

<http://muse-test.c2dh.uni.lux:8080>

KEYWORD SEARCH powered by Solr

You have to write at least 3 letters to search different sources.

AWS_RAG solution

<https://d-936791057c.awsapps.com/start/>



AWS account creation



Hello Tugce Karatas,

Your administrator for AWS Account #918687030054 has invited you to AWS IAM Identity Center. Accepting this invitation activates your user account in IAM Identity Center so that you can access assigned AWS resources. Choose the link below to accept this invitation.

Accept invitation

This invitation will expire in 7 days.

Accessing the AWS access portal

After you've accepted the invitation, you can sign in to the AWS access portal by using the information below.

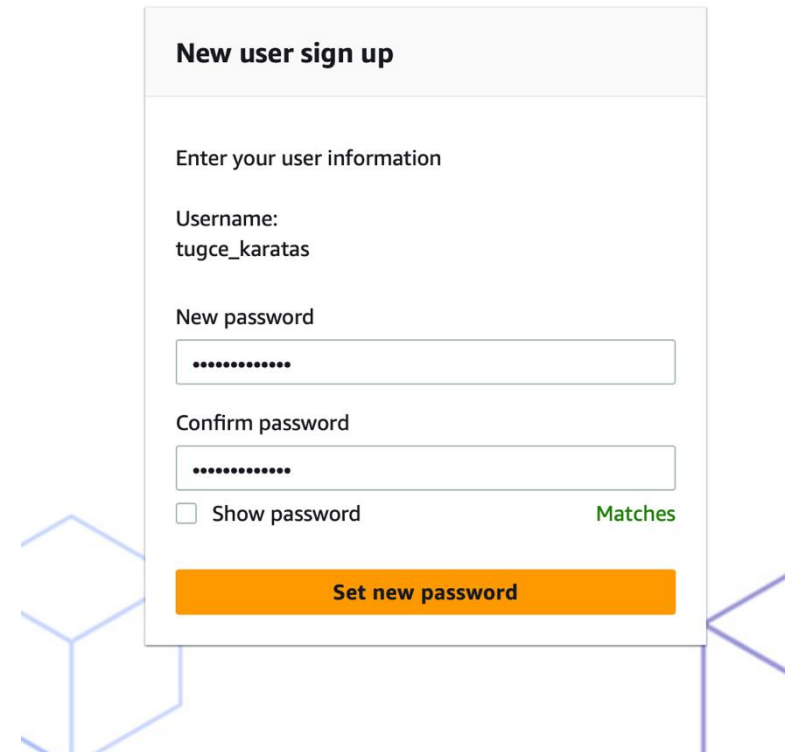
Your AWS access portal URL:

<https://d-936791057c.awsapps.com/start/>

Your Username:

tugce_karatas

1- accept the email invitation



The form is titled "New user sign up" and is used for creating a new AWS account. It contains the following fields and elements:

- Enter your user information**
- Username:** tugce_karatas
- New password:** A text input field with a masked password (dots).
- Confirm password:** A text input field with a masked password (dots).
- ☐ Show password
- Matches** (green text)
- Set new password** (orange button)

2- create an account and choose password

AWS account creation

Register MFA device

Username:
tugce_karatas (not you?)

Your organization requires multi-factor authentication (MFA) for added security during sign-in. Each time you sign in, you'll be prompted for your password and an MFA device.
[Learn more](#)

Select one of the options below to get started:

☒  **Authenticator app**
Authenticate using a code generated by an app installed on your mobile device or computer.

☐  **Security key**
Authenticate by touching a hardware security key such as YubiKey, Feltian, etc.

☐  **Built-in authenticator**
Authenticate using a fingerprint scanner or camera built-in to your computer such as Apple TouchID, Windows Hello, etc.

[Next](#)

3- Authenticate the account using the app
(the same as university one)

Set up the authenticator app

Username:
tugce_karatas (not you?)

[Back to MFA device options](#)

1



Install either the Google Authenticator, Duo Mobile, or Authy app on your mobile device or computer. [See a list of compatible apps](#)

2



Use your virtual MFA app or your device's camera to scan the QR code. [Show secret key.](#)

3

Enter the six digit code from your authenticator app

Authenticator code

4- choose QR code option with and scan the code

AWS account creation



Authenticator app registered

✓ Your authenticator app has been successfully registered. You can now use it when prompted for additional verification at sign in.

tugce_karatas's MFA 1 [Rename](#)

Type and description: Authenticator app

Done

5- you should see this confirmation screen

aws access portal

AWS access portal

Accounts

Applications

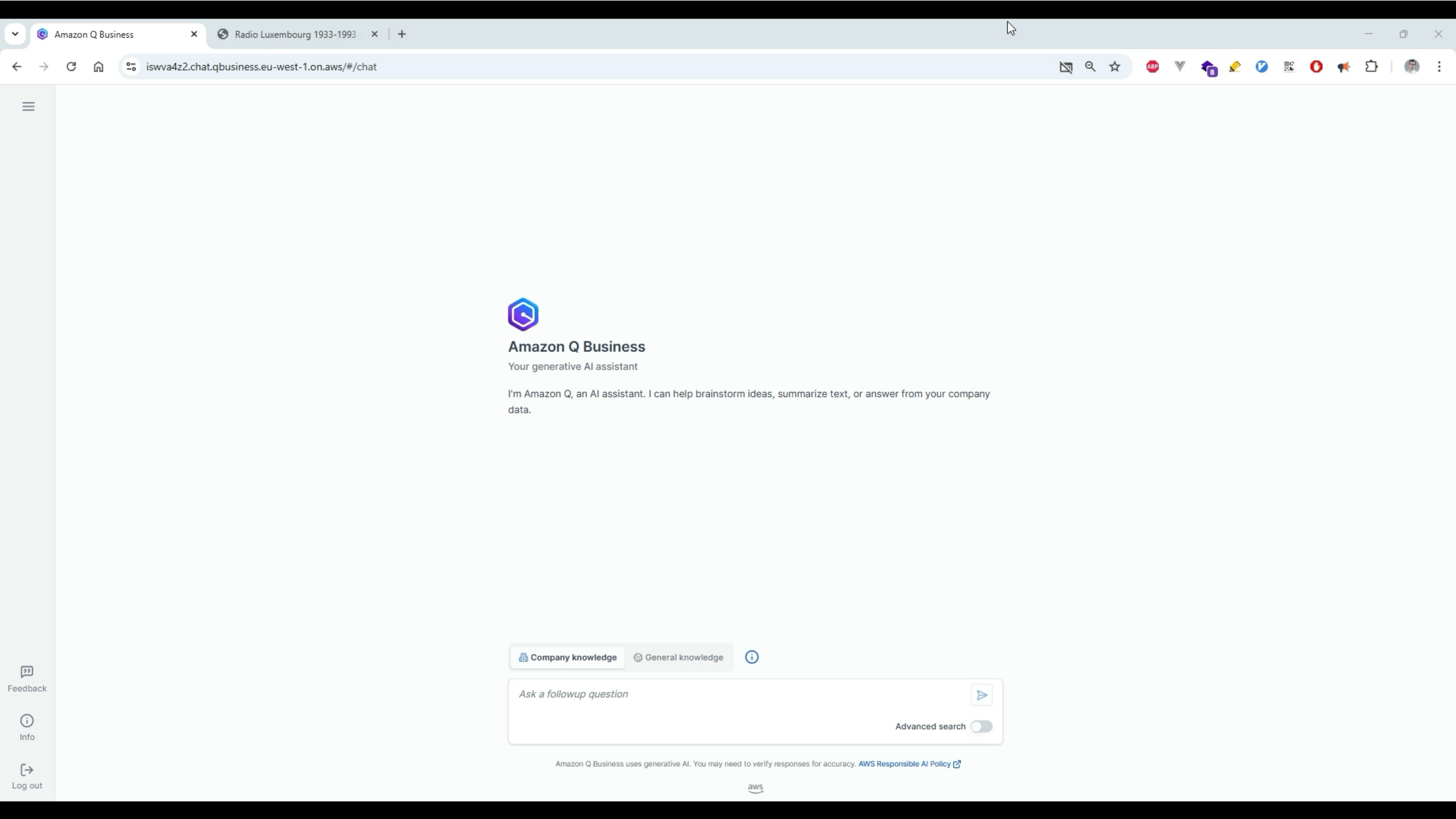
Applications (1)

Find applications by name

QBusiness-App-slod-prod

QBusiness-App-slod-prod

6- you are in the portal you can click the Qpod app



Amazon Q Business

Your generative AI assistant

I'm Amazon Q, an AI assistant. I can help brainstorm ideas, summarize text, or answer from your company data.

Company knowledge

General knowledge



Ask a followup question



Advanced search



Amazon Q Business uses generative AI. You may need to verify responses for accuracy. [AWS Responsible AI Policy](#)



AWS RAG Understands natural language

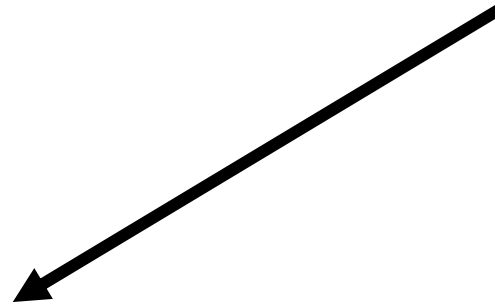
Query: “Quote Polish politicians on Radio Luxembourg”



Interpreted correctly as “Please report what Polish politicians said with regard to their experiences with Radio Luxembourg”



Skip part of the book where a phrase “Polish politicians” appears, but in the context of experiences of Ambassador of Luxembourg in Poland



Find quotes from:

- “former First Lady of the Republic of Poland”,
- “former Minister of Foreign Affairs” (of Poland, which is not mentioned directly),
- director of “Institute of National Remembrance Branch in Warsaw”.

And assign correct quotations from them, not the text that is right after their names.

In October 2010 I came to Warsaw as the new Ambassador of the Grand Duchy of Luxembourg to the Republic of Poland. During my first courtesy calls to Polish politicians, high-level civil servants and business people I was struck by the fact that so many of my interlocutors spontaneously associated my country first and foremost with *Radio Luxembourg*. Indeed, many representatives of modern Poland’s political, economic and cultural elites regularly listened to the English-language programmes of *Radio Luxembourg* in their youth. By chance, soon after my arrival in



Quote Polish politicians on Radio Luxembourg



Several Polish politicians and prominent figures have shared their experiences with Radio Luxembourg:

Andrzej Olechowski, economist and former Minister of Foreign Affairs, credited Radio Luxembourg as a key motivator for learning English. He stated that listening to the station and its music became an incentive to learn the language, which he considers one of the most important decisions in his life. ^[1]

Jolanta Kwaśniewska, Former First Lady of the Republic of Poland, recalled the challenges of trying to find the best reception for Radio Luxembourg. She mentioned moving around rooms with friends, carrying the radio set to find the optimal spot for listening. ^[1]

Maria Szablowska, a music journalist, noted the widespread popularity of Radio Luxembourg, stating that virtually everyone in Poland listened to it, especially high school and university students. ^[1]

Prof. Jerzy Eisler, Director of the Institute of National Remembrance Branch in Warsaw, observed that Radio Luxembourg was primarily listened to by wealthier students in cities who had access to decent radio sets. He described the station as a "window onto the world" for these listeners. ^[1]

These quotes highlight the significant impact Radio Luxembourg had on Polish listeners, particularly in terms of language learning, cultural exposure, and as a source of Western music during the communist era.

[Sources](#)



^[1] [Remembering Radio Luxembourg in the Peoples Republic of Poland.pdf](#)

[https://s3-datasource-slod-prod.s3.eu-west-1.amazo...%20in%20the%20Peoples%20Republic%20of%20Poland.pdf](https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/in%20the%20Peoples%20Republic%20of%20Poland.pdf)

Updated November 7, 2025

Example: Remembering Radio Luxembourg in the People's Republic of Poland

Multilingual



Quand Radio Luxembourg a-t-elle été fondée ?



Radio Luxembourg a commencé ses émissions le 15 mars 1933. [1] Cette date marque le début réel de l'existence de la station. [1] Les auditeurs à travers l'Europe ont pu entendre pour la première fois le générique familier "Bonjour, bonjour, ami fidèle, c'est Luxembourg, c'est Luxembourg, Bonjour, bonjour, c'est Luxembourg qui vous appelle !". [1]

Cependant, il est important de noter que les origines de Radio Luxembourg remontent un peu plus tôt. En 1929, une entreprise appelée Compagnie Nationale de Radiodiffusion Luxembourgeoise (émissions Radio-Luxembourg) a été créée. [1] Cette société avait pour objet la construction et l'exploitation de postes de radiodiffusion. [1]

De plus, dès 1925, un commerçant nommé François Anen avait créé une association appelée Radio-Luxembourg et mis en place un poste d'une puissance de 50 watts. [1] En 1926, la puissance de ce poste a été augmentée à 150 watts. [1]

▼ Events

▼ Sources



Wann wurde Radio Luxembourg gegründet?



Radio Luxemburg wurde offiziell am 15. März 1933 gegründet. [1] An diesem Tag begann die Station tatsächlich zu existieren und sendete für Hörer in ganz Europa. [1] Die Planungen für die Gründung einer kommerziellen Radiostation in Luxemburg begannen jedoch schon früher. Bereits 1929 wandten sich zwei französische Geschäftsleute an die luxemburgische Regierung, um die Erlaubnis für den Start eines Radiosenders zu erhalten. [2] Sie erhielten eine Lizenz, und die ersten Sendungen gingen zwei Jahre später, also 1931, auf Sendung. [2] Von Anfang an verfolgte der Sender einen internationalen Ansatz und strahlte experimentelle Programme auf Französisch, Deutsch und Luxemburgisch für einheimische Hörer und über die Grenzen in die Nachbarländer aus. [2]

Möchten Sie weitere Informationen zu einem bestimmten Aspekt der Gründungsgeschichte von Radio Luxemburg, wie zum Beispiel die Vorbereitungen vor der offiziellen Gründung oder die frühen Sendungen?



When was Radio Luxembourg founded?



Radio Luxembourg was officially founded on March 15, 1933. [1] This launch came after a group of French visionaries received a license from the Luxembourg government on December 29, 1930, to operate a commercial radio station in Luxembourg. [1]

The journey to establish Radio Luxembourg began earlier, with the first broadcasts starting in the 1920s. In 1925, François Anen, a merchant, created an association to set up a small 50-watt transmitter. [1] The following year, the power was increased to 150 watts. [1]

However, the full-fledged commercial radio station that became Radio Luxembourg came into existence in 1933. [1] This marked the beginning of its long presence in European broadcasting, despite a six-year interruption during World War II. [1]

▼ Events

▲ Sources



[1] [Radio Luxembourg 1933-1993.pdf](#)

<https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/audio-luxembourg/Radio%20Luxembourg%201933-1993.pdf>

Updated November 7, 2025

[2] [The station of the stars - Radio Luxembourg - 1965.pdf](#)

<https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/he%20stars%20-%20Radio%20Luxembourg%20-%201965.pdf>

Updated November 7, 2025

[Radio Luxembourg 1933-1993.pdf](#)

<https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/audio-luxembourg/Radio%20Luxembourg%201933-1993.pdf>

“ univers de l’audiovisuel, Radio Luxembourg s’apparente à un monument par le rayonnement de son audience d’abord, par sa longévité ensuite. A l’aune de la radiodiffusion, son histoire appartient à la longue durée malgré une interruption de six années lors de la Seconde Guerre...”

Permission-based access

If a user does not have permissions to view a document, the system will refuse to retrieve information.
Here, two different users get a different answers based on their access rights:

The image displays two side-by-side chat interface screenshots, illustrating permission-based access. Both screenshots show a user asking the question: "What are the strategic planning assumptions for the year 2025?".

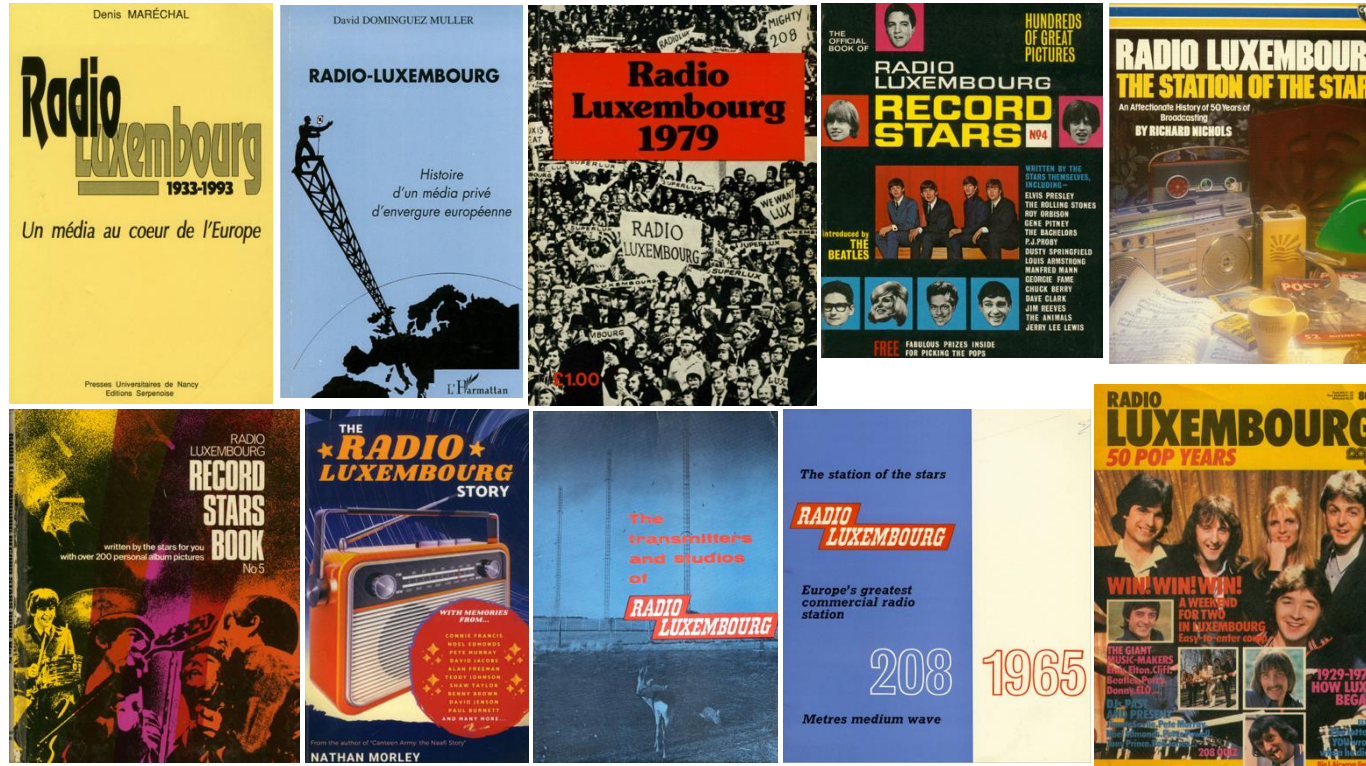
Left Screenshot (User without permissions):

- The chat bubble shows a response: "Sorry, I could not find relevant information to complete your request."
- Below the response is a button labeled "No sources".
- At the bottom right of the chat bubble are three icons: a thumbs up, a thumbs down, and a share icon.

Right Screenshot (User with permissions):

- The chat bubble shows a response: "The strategic planning assumptions for the year 2025 according to your source data are:"
- Below this, there are two bullet points:
 - By 2025, 50% of data scientist activities will be automated by AI, easing the acute talent shortage. [1]
 - In addition, I also found the following possible assumption:
By 2023, 75% of all databases will be on a cloud platform — a development that will drastically change the DBMS vendor landscape. [2]
- Below the response is a button labeled "Sources" with a dropdown arrow.
- At the bottom right of the chat bubble are three icons: a thumbs up, a thumbs down, and a share icon.
- Below the chat bubble, there are two numbered links:
 - [1] [MQAIDeveloperServices.pdf](https://demo.s3.amazonaws.com/Data/analyst-reports/MQAIDeveloperServices.pdf)
 - [2] [MQOperationalDBMS.pdf](https://demo.s3.amazonaws.com/Data/analyst-reports/MQOperationalDBMS.pdf)

Permission-based access. Example data

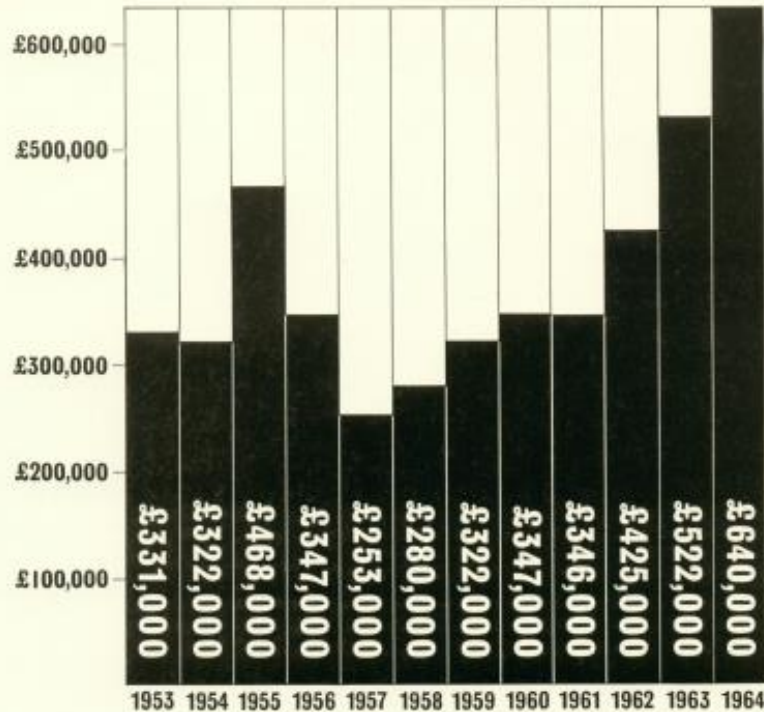


Whole DRI team has access



Only members of the private project have access

AIRTIME SALES 1953 - 1964



AIRTIME RATES WERE INCREASED
IN 1964 BY APPROXIMATELY 11%

Charts are interpreted too!



What were airtime sales of Radio Luxembourg in '50 and '60?



Radio Luxembourg's airtime sales figures for the 1950s and 1960s show significant fluctuations over the years. In 1953, the sales were £331,000, which increased to £468,000 in 1955. [1] The sales then experienced a decline, reaching £253,000 in 1957. [1] As the decade progressed, there was a gradual increase in airtime sales. By 1960, the sales had risen to £347,000. [1] The upward trend continued into the early 1960s, with sales reaching £425,000 in 1962, £532,000 in 1963, and peaking at £640,000 in 1964. [1] This data demonstrates a general growth in Radio Luxembourg's airtime sales from the 1950s to the 1960s, with the latter part of the 1960s showing particularly strong performance.

▼ Events

▲ Sources



[1] 📄 The station of the stars - Radio Luxembourg - 1965.pdf 🔗

🌐 <https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/he%20stars%20-%20Radio%20Luxembourg%20-%201965.pdf>

Updated November 7, 2025



THE RADIO LUXEMBOURG AUDIENCE

Radio listening, both in the United States and in this country, has made a remarkable recovery during the past ten years. Not so many years ago radio stations in the States were going for a song and then they started being

Example: Radio Luxembourg - The station of the stars. 1965.

Pricing



- **Each account costs \$3 per month**
- **Consumption price:**
 - **\$200 for 30,000 units/month** (but what is a unit?)

Starter Index

The **Starter Index** is deployed in a single Availability Zone, making it ideal for PoCs and developer workloads.

- Priced at \$0.140 per hour for one unit (limit five units per application)
- \$0.003 per image (\$3/1000 images)
- Audio \$0.006/minute
- Video \$0.050/minute
- One index unit includes:
 - 100 hours of connector usage per month
 - 20,000 documents or 200 MB of extracted text, whichever comes first

Enterprise Index

The **Enterprise Index** is deployed across three Availability Zones, making it best for production workloads.

- Priced at \$0.264 per hour for one unit
- \$0.003 per image (\$3/1000 images)
- Audio \$0.006/minute
- Video \$0.050/minute
- One index unit includes:
 - 100 hours of connector usage per month
 - 20,000 documents or 200 MB of extracted text, whichever comes first

Free Trial terms	
Amazon Q Business user subscriptions (Amazon Q Business Pro and Lite)	Amazon Q Business offers a 60-day free trial for up to 50 Amazon Q Business Pro or Lite users per application
Amazon Q Business Index (Enterprise and Starter)	Free trial of 1,500 index hours per application to be used in 60 days
Amazon Q in QuickSight user subscription (Amazon Q Business Pro)	30-day free trial for up to 4 users per Amazon QuickSight account

COMPARISON

	MuSE	RAG
Hosted Solution	Internal	AWS
Locally Installed	Yes	No
Open Source/Proprietary	Yes	No
Customer Support	No	Yes (paid)
System Upgrades	Manual	Automatic
Cost	Fixed	Usage-based
Implementation	Development required	Off the shelf
User friendly	UX required	Intuitive
Privacy	Institutional level	Researcher level

NEXT STEP

Scenario 1 — Outsourcing AI Practices

- Identify external providers that match your domain needs.
- Define requirements: data types, workflows, success indicators.
- Establish data-sharing and privacy agreements.
- Plan for risk mitigation around vendor dependency and continuity.

Scenario 2 — In-House AI Practices

- Assess internal expertise, staffing gaps, and resource needs.
- Build or expand internal data/AI team.
- Begin designing internal workflows, pipelines, and evaluation standards.
- Budget for infrastructure, tools, and long-term maintenance.

Scenario 3 - Institutional Data Strategy (Hybrid Approach)

- Map which functions will be outsourced vs. handled internally.
- Define shared responsibilities between internal teams and external services.
- Create unified data governance framework.
- Begin coordination planning and workflow alignment.